

Evaluating Weakly Supervised Cryptocurrency Risk-State Learning: When Label Fit, Stability, and Semantics Disagree

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ABSTRACT

Weak supervision is increasingly used for market-state learning when regime labels are unavailable, but what heuristic proxy labels actually transmit to learned state representations remains unclear. We study that question on Binance spot 1-hour cryptocurrency data using past-only proxy labels built from return, volatility, moving-average-gap, and drawdown signals. We formalize a three-axis audit protocol separating proxy-label agreement, temporal stability, and ex-post state semantics, with the semantic axis screened by nonparametric tests and a block-bootstrap robustness check. Across assets and threshold variants, learned states align more consistently with future risk stratification than with return direction, and model rankings reverse once evaluation shifts across fit, smoothness, and semantics. On BTCUSDT, TCN-96x4 improves proxy-label macro-F1 over the default GRU, yet switches more often and is weaker on nominal significant risk layering. HistGradientBoosting chosen by weighted classification-plus-stability scoring and TOPSIS loses 6/21 nominal significant-risk-layering checks relative to the semantic selector before those gaps compress under dependence-aware resampling. Tested semantic-regularization variants show no measurable gain, while a Semantic Orientation Ratio isolates XRP as the only structurally misaligned asset. Semantic alignment with persistent risk is therefore achievable, but it is driven more by proxy-label design than by training objectives.

1. Introduction

Weak supervision provides a practical training signal when exact labels are unavailable, but it raises a broader question: how much of the deployment objective survives the translation from heuristic proxy labels to learned representations? In financial time-series learning, this question is especially sharp because the proxy labels that are easy to construct from market observables, such as trend indicators, volatility thresholds, and drawdown rules, may carry systematically different information than the downstream states users actually need. Cryptocurrency risk-monitoring, regime-filtering, and automated trading systems increasingly rely on compact market-state models, yet those models are still commonly selected by how well they reproduce the heuristic itself before anyone checks whether the resulting state path is stable and operationally meaningful. Figure 1 summarizes the resulting weak-supervision and evaluation workflow. Regime-switching and hidden-state models provide a long-established language for this problem in macroeconomics and finance [1–4].

This paper studies the question empirically in cryptocurrency market-state learning. Using hourly Binance spot candlestick data and heuristic proxy labels constructed from return, volatility, moving-average-gap, and drawdown signals, we ask whether learned state representations inherit the risk-stratification structure latent in the proxy labels, or converge to something else. The goal is not open-ended

state discovery, but a compact and deployment-oriented state path that distinguishes persistent market conditions reproducibly under modest computing resources. Recent work has applied related ideas to bull/bear detection, multi-state market classification, and machine-learning-based regime recognition in financial markets [5–8]. In cryptocurrency markets, related studies report regime effects in joint return dynamics, volatility, illiquidity, and tail risk [9–12]. Our scientific question is narrower and more general than a reproducibility exercise: when supervision is heuristic, noisy, and only partially aligned with latent regimes of interest, what exactly can weak supervision learn, and what determines the boundary of that learning?

The central difficulty is that hourly market states do not come with clean ground-truth labels. Pure rule systems are easy to reproduce but tend to be brittle. Purely unsupervised state discovery is flexible but often drifts away from the decision question that users actually care about. Weak supervision and noisy-label learning provide a useful middle ground when the available supervisory signal is heuristic, partial, or corrupted rather than exact [13–18]. In this setting, the target should not be the heuristic label itself, but the latent persistent regime that the heuristic label only imperfectly reveals. We therefore construct coarse proxy labels from past-only trend, volatility, and drawdown statistics, and train temporal models to distill these noisy local cues into a more stable state representation.

The central finding is that weak supervision reliably recovers risk stratification but not return direction, and that this asymmetry is structurally determined by the information content of proxy labels rather than by training objectives. To establish this finding rigorously, we formalize a three-axis evaluation protocol that separates proxy-label agreement,

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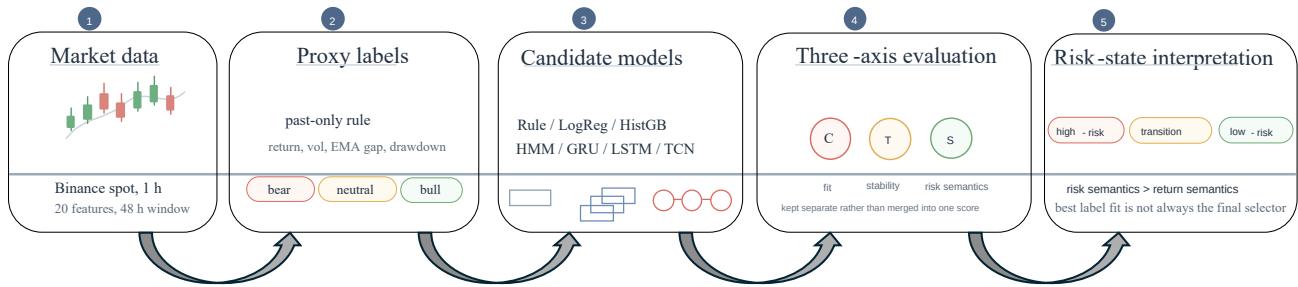


Figure 1: Overview of the weakly supervised risk-state learning and evaluation workflow. Market windows are converted into rule-based weak labels, candidate learners are evaluated along proxy-label fit, temporal stability, and ex-post risk semantics, and the reported state path is interpreted together with possible cross-axis ranking reversals.

temporal stability, and ex-post state semantics; within the semantic axis, it further distinguishes raw ordering, nominal significant risk layering, and block-bootstrap robustness. Across multi-asset tests, temporal-backbone comparisons, stronger static baselines, deconfounding checks, ablations, overlap controls, and threshold sensitivity analyses, the experiments show that these axes do not induce the same ranking, and that nominal semantic gaps can contract materially once serial dependence is respected more explicitly. The protocol is therefore not the endpoint of the paper, but the measurement instrument that makes visible what weak supervision has and has not learned in a deployment-oriented risk-state setting.

2. Related Work

2.1. Market-State and Regime Identification

Classical regime-switching research models structural change through latent states and state-dependent dynamics [1–4]. This literature established the vocabulary that still shapes market-state modelling in finance, especially the emphasis on latent regimes, transition probabilities, persistence, and state-conditioned moments. Building on this foundation, later work studied bull/bear identification and broader multi-state market classification through dynamic binary models, hierarchical hidden Markov models, long-horizon taxonomies, and generative-discriminative classifiers [5–8]. These studies are close to our decision problem in spirit, but most focus on equity markets or directional market classification.

Within cryptocurrency markets, related research has documented regime structure in joint return dynamics, Bitcoin volatility, illiquidity, spillovers, and tail risk, often through Markov-switching or regime-dependent risk models [9–12, 19–26]. More recent machine-learning studies extend this line toward crypto volatility forecasting, driver analysis, and broader reviews of the volatility literature [27–29]. Most of this work targets volatility prediction, value-at-risk, cross-asset dependence, or directional forecasting rather than learning a stable three-state path directly from spot K-line windows. Our objective is therefore narrower and

more operational: to infer whether the resulting states behave as persistent risk strata, not merely as short-horizon return signals.

2.2. Weak Supervision, Noisy Labels, and State Persistence

Weak supervision and noisy-label learning provide a natural framework for training from heuristic, partial, or corrupted labels rather than exact annotations [13–18, 30–32]. This framing matches our setting because the supervisory signal is not hand-labeled ground truth, but a reproducible proxy rule built from past-only return, volatility, moving-average-gap, and drawdown statistics. Recent financial time-series work on adaptive label correction points in the same direction by showing that hand-designed labels can be useful while also making their thresholds, horizons, and correction rules part of the modelling problem [33]. In that sense, the proxy labels used here should be read as imperfect supervision that may reveal a latent regime only noisily.

A separate but complementary requirement is that the inferred state path should be temporally persistent rather than switching unrealistically often [34–37]. For market-state models, persistence is not a cosmetic smoothing preference; it is part of what makes the states interpretable and usable as a regime representation. The combination of these two ideas motivates our training target: the model should distill a latent, temporally stable state sequence from heuristic labels rather than merely replicate the label rule bar by bar.

A less-studied question is what information heuristic labels can reliably transmit to learned representations. In weakly supervised pipelines, the gap between proxy-label objectives and deployment objectives can be systematic rather than random, especially when the proxy labels compress several economic signals into a single discrete heuristic. In time-series settings this gap is compounded by temporal dependence, state imbalance, and overlapping windows, which can cause models to learn the noise structure of proxy labels more faithfully than the latent state dynamics those labels are meant to approximate. The present study addresses

this question directly by testing whether learned states inherit the risk-stratification structure of proxy labels, and by identifying the conditions under which that inheritance fails.

2.3. Model Evaluation and Temporal Architectures

Conventional multi-criteria model selection aggregates observed metrics through weighted sums, distance-to-ideal ranking, or related multi-attribute decision rules such as TOPSIS [38]. These methods are useful once the evaluation criteria are already accepted as meaningful objectives. In weakly supervised regime learning, however, the prior question is whether proxy-label fit, temporal smoothness, and ex-post semantics are even commensurable enough to be collapsed into one score. A model can score well on proxy-label agreement and temporal smoothness, yet still produce states whose ex-post economic meaning is weak or unstable. Our evaluation protocol should therefore be read less as a new generic multi-criteria optimizer than as an audit layer that retains conventional classification and stability metrics while adding explicit semantic checks for whether the selected state path behaves like a risk-state taxonomy after the fact. Section 6.4 quantifies this increment against classification-only, weighted classification-plus-stability, and TOPSIS-style classification-plus-stability selectors, and shows that the added value appears precisely when stronger label-fitting candidates make scalar aggregation misleading.

Methodologically, the temporal neural baselines used here are standard rather than architecturally novel. LSTM and GRU encoders remain canonical recurrent models for learning medium- and long-range temporal dependencies [39, 40], while temporal convolutional networks provide a non-recurrent alternative based on causal, dilated convolutions and residual blocks that has been shown to compete well on generic sequence modelling tasks [41]. Transformer-style time-series backbones such as PatchTST and iTransformer have become influential in forecasting and representation learning [42, 43]. We therefore include a lightweight two-layer TransformerEncoder stress test on BTCUSDT, but the present study still uses these backbones only to stress-test the evaluation protocol rather than to claim architectural novelty. The experiments therefore focus on whether stronger temporal feature extractors improve only proxy-label fit or also the stability and risk semantics of the inferred state path.

3. Problem Formulation, Data, and Weak Labels

3.1. Problem Formulation

We formulate hourly cryptocurrency market-state identification as a weakly supervised three-class sequence labeling problem, where the learning target is a temporally persistent latent risk-state path rather than exact replication of heuristic proxy labels. Let $x_t \in \mathbb{R}^d$ denote the feature vector extracted from the 1-hour candlestick at time t , and let $X_t = [x_{t-w+1}, \dots, x_t]$ be the rolling window of length w .

The objective is to estimate a three-state regime variable

$$z_t \in \begin{cases} \text{high-risk,} \\ \text{transition,} \\ \text{low-risk,} \end{cases}$$

with outputs represented as class probabilities

$$\begin{aligned} P(z_t = \text{high-risk}), \\ P(z_t = \text{transition}), \\ P(z_t = \text{low-risk}). \end{aligned}$$

The learning target is not a manually labeled ground truth. Instead, we construct a proxy label $\tilde{z}_t$ from past-only return, moving-average gap, volatility, and drawdown signals. Throughout the manuscript, the labels bear, neutral, and bull are preserved for readability; their directional interpretation is treated as an empirical hypothesis rather than a modelling assumption. The learning problem is therefore weak supervision [13–15]: the model should distill a latent, temporally persistent regime path from $\tilde{z}_t$ rather than merely memorize the rule that generated it.

3.2. Data Source and Split

The experiments use Binance spot 1-hour candlestick data from 2021-01-01 to 2026-04-20. BTCUSDT and ETHUSDT are the main temporal-model benchmark assets; BNBUSDT, SOLUSDT, and XRPUSDT are added as a multi-asset robustness set for the weak-label, static, post-processed static, and cross-asset-transfer analyses. The default chronological split is:

- Training: 2021-01-01 to 2023-12-31.
- Validation: 2024-01-01 to 2024-12-31.
- Test: 2025-01-01 to 2026-04-20.

All splits are chronological. No random reshuffling is used. The test span is reported both as a whole and by the subperiods 2025-H1, 2025-H2, and 2026-YTD (2026-01-01 to 2026-04-20), because a pooled 16-month result can hide whether the learned semantics survive changing market conditions inside the holdout period.

The three BTCUSDT test subperiods also cover materially different market backgrounds. Using the close-to-close path of the underlying spot series, BTCUSDT gains 13.5% in 2025-H1, then loses 18.4% in 2025-H2 and 13.6% in 2026-YTD, while the corresponding max drawdowns remain severe at roughly -30.9%, -34.8%, and -35.6%. This mix of rebound and drawdown phases is one reason why a single pooled directional-return interpretation is unstable even when ex-post risk ordering remains comparatively persistent.

Across the chronological splits, the default proxy labels are imbalanced. For the BTCUSDT/ETHUSDT benchmark assets, neutral occupies 74.7%–81.3% of observations, bear 10.6%–22.0%, and bull 3.3%–10.0% across train, validation,

and test segments. The added assets make the imbalance more visible: under the default rule, SOLUSDT and XRPUSDT have sparse bull states in some splits, and under the strict threshold-sensitivity setting their test bull counts fall to 17 and 14 bars, respectively. This imbalance is one reason why macro-F1, balanced accuracy, semantic ordering, significance checks, and explicit check-coverage flags are reported instead of accuracy alone.

3.3. Feature Set

Each hourly bar is transformed into 20 hand-crafted features derived from candlestick fields: log returns over 1, 4, and 24 hours; rolling volatility over 24 and 72 hours; high–low range; open–close change; 24-hour z-scores of volume, quote volume, and trade count; exponential moving-average gaps over 12–48 and 24–72 hours; 24-hour rolling skewness and kurtosis; 24-hour up-bar and down-bar ratios; 72-hour maximum drawdown; 24-hour trend strength; taker-buy ratio; and 24-hour volume–price correlation. A 48-step rolling window is then formed for the temporal models. The inclusion of `max_drawdown_72` deserves advance notice because the same statistic also appears in the weak-label rule below. It is therefore an explicit modelling anchor, not a neutral feature, and later deconfounding checks are used to show how much of the recovered risk semantics depend on that anchor.

3.4. Proxy Label Construction

The weak labels are constructed from past-only statistics. Historically, the rule generator emits the names bull, bear, and neutral. In the current implementation, a bar is labeled bull when

$$\begin{aligned} \log_return_{24} &> 0.01, \\ \log_return_{24} &> \text{rolling_vol}_{24}, \\ \text{ema_gap}_{24_72} &> 0.002, \\ \text{max_drawdown}_{72} &> -0.03. \end{aligned}$$

It is labeled bear when

$$\begin{aligned} \log_return_{24} &< -0.01, \\ \log_return_{24} &< -\text{rolling_vol}_{24}, \\ \text{ema_gap}_{24_72} &< -0.002, \\ \text{max_drawdown}_{72} &< -0.06. \end{aligned}$$

Remaining periods are assigned to neutral. This construction avoids forward-looking leakage in the training target. It also reuses the drawdown statistic on both the feature side and the label side, so the downstream taxonomy is partly anchored to observable downside stress instead of being discovered in a fully free-form way. The labels bear, neutral, and bull are preserved for readability; their directional interpretation is treated as an empirical hypothesis rather than a modelling assumption. Later semantic analysis shows why those names need to be treated cautiously: empirically, the learned states align more consistently with future risk stratification than with future return direction.

4. Evaluated Temporal Models and Training Objective

The evaluated temporal models receive a rolling window X_t and output three-class regime probabilities. The default recurrent variant is a single-layer GRU encoder followed by LayerNorm, dropout, and a linear classification head. We also evaluate a vanilla LSTM under the same windowing and optimization protocol, and a causal dilated TCN with residual temporal convolution blocks. These encoders are treated as comparable backbones inside the same weak-supervision pipeline, not as separate problem formulations.

In addition to cross-entropy on the proxy labels, the temporal models can use two auxiliary regularization terms:

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}. \quad (1)$$

The class-balance term penalizes degenerate solutions that collapse into a single dominant class. The continuity term discourages excessive probability changes between adjacent windows:

$$\mathcal{L}_{\text{cont}} = \frac{1}{T-1} \sum_{t=2}^T g_t \left\| \text{softmax}(f(X_t)) - \text{softmax}(f(X_{t-1})) \right\|_1, \quad (2)$$

where g_t is a volatility-aware gate,

$$g_t = \frac{1}{1 + \alpha \bar{v}_t}, \quad (3)$$

and $\bar{v}_t$ is the local rolling volatility summary. Low-volatility periods therefore receive stronger smoothing, while high-volatility periods retain more freedom to change state quickly. In the current implementation, this objective is applied as a chronological batch-local approximation over adjacent windows inside ordered batches rather than as a strict sequence-global penalty. It should therefore be read as a local ordered-window smoothness bias, not as an exact full-path regularizer. The empirical value of these auxiliary terms is examined later by ablation and overlap controls rather than assumed a priori.

5. Experimental Setup and Evaluation Protocol

5.1. Baselines

The comparison uses a compact but stronger baseline set:

- Rule-based proxy reference: directly reuse the proxy labels as predictions.
- Logistic regression: a static supervised baseline on the hand-crafted features.
- Logistic regression with temporal post-processing: rolling probability smoothing plus a switch-margin rule. This is the key low-cost baseline because it directly tests whether a static classifier with simple smoothing is already sufficient.

- **HistGradientBoosting**: a stronger non-linear static baseline on the same hand-crafted features, reported both in raw form and with the same temporal post-processing used for logistic regression.
- **Gaussian HMM**: a classical temporal state baseline with three latent states first mapped back to the proxy-label space by majority voting on the training split, then re-examined under an ex-post semantic ordering diagnostic to separate concept-drift-sensitive label mapping from the latent path's own realized risk ordering [1, 2, 5].
- **Vanilla LSTM and TCN**: standard temporal backbones evaluated under the same feature set, chronological split, and loss protocol as the GRU [39–41].
- **Lightweight Transformer encoder**: a BTCUSDT-only stress-test reference included to check whether an attention-style temporal backbone overturns the main trade-off [42, 43].

The rule-based line should be interpreted carefully: because it directly reuses the proxy labels, it is a consistency reference rather than a learned competitor. The neural temporal backbones share the same data pipeline and 48-hour window. Multi-seed checks are used later to determine whether the ranking between post-processed static baselines and recurrent encoders is stable or merely a single-seed accident.

5.2. Implementation Details

The temporal models use 48-step windows and the 20-feature input described above. GRU and LSTM use one recurrent layer with hidden dimension 64 and dropout 0.1, followed by LayerNorm, dropout, and a linear classification head. The TCN uses causal convolutions with residual blocks; the reported stress test includes 64-by-3, 128-by-3, and 96-by-4 hidden-dimension/layer configurations. The lightweight Transformer stress test uses two encoder layers with hidden dimension 64, four attention heads, feed-forward dimension 256, sinusoidal positional encoding, and the same dropout 0.1. It is included as a deliberately simple attention-style reference rather than as a state-of-the-art Transformer benchmark: no patching, channel-independent tokenization, or architecture-specific pretraining is used. Training uses AdamW with learning rate 10^{-3} , weight decay 10^{-4} , batch size 256, and 12 epochs. Unless otherwise stated, the checkpoint with the best validation macro-F1 is retained for testing. For the main GRU, the continuity weight is $\lambda_{\text{cont}} = 0.15$, the class-balance weight is $\lambda_{\text{bal}} = 0.05$, and the gate strength is $\alpha = 3.0$. The temporal post-processing baseline uses a 6-bar rolling mean with a switch margin of 0.10. Main-result tables report the seed-42 run, and separate five-seed summaries for GRU, TCN-96x4, and Transformer-2L are used as robustness checks. A later checkpoint-selection diagnostic compares this default against a semantic-aware alternative, but the main benchmark still uses the label-fit-driven selector so that the comparison with earlier baselines

remains fixed. The main text reports seed-42 for consistency, whereas seed-based robustness summaries are used to bound how much both classification and semantic conclusions move across repeated training runs.

5.3. Evaluation Metrics

The evaluation protocol is organized around three categories of metrics. One category covers classification against the proxy labels: macro-F1, balanced accuracy, bull F1, and bear F1. These measure agreement with the heuristic supervisory signal, not direct accuracy against latent ground-truth regimes. A second category covers temporal stability in the predicted sequences through the total number of regime switches, the average state duration, and the daily switching rate. The third covers ex-post semantics, where we test whether the learned states align more consistently with future return ordering or with future risk ordering. We use cross-method ranking reversal for cases where two methods exchange order across evaluation axes, and cross-axis inconsistency for cases where one method looks strong on one axis but weak on another. For that reason the protocol does not collapse these axes into a single utility score.

Formally, for asset a and method m , the protocol returns

$$\Pi(m, a) = (C(m, a), T(m, a), S(m, a)),$$

where $C(m, a)$ is the proxy-label agreement vector, $T(m, a)$ is the temporal-stability vector, and

$$S(m, a) = (r^{\text{risk}}, r^{\text{return}}, g^{\text{nom}}, g^{\text{boot}})$$

collects raw risk-order consistency, raw return-order consistency, nominal significant-risk layering, and, when computed, block-bootstrap robustness. The protocol remains multi-objective: it does not aggregate these quantities into a single score because the experiments repeatedly show that such aggregation would hide cross-method ranking reversals and within-method cross-axis inconsistencies. Operationally, the protocol should be read as both an audit filter and a measurement instrument. A model that dominates on proxy-label fit but fails semantic checks is still informative about label learnability, but it is not an equally credible deployment choice. Conversely, separating proxy-label agreement, temporal stability, and ex-post semantics makes it possible to answer the paper's central scientific question: what weak supervision has learned, what it has not learned, and which evaluation axis reveals each part of that boundary.

For each default asset–method pair, risk-order consistency is computed over 21 checks: seven risk-oriented metrics (future volatility, absolute return, path loss, path gain, path range, and two downside-hit rates) evaluated over the three test subperiods 2025-H1, 2025-H2, and 2026-YTD. A risk check is counted as matched when the descending state means satisfy bear > neutral > bull. Return-order consistency is computed over six checks using future 24-hour and 72-hour returns with the expected order bull > neutral > bear. To move beyond descriptive ordering, we also run

Kruskal–Wallis tests and one-sided Mann–Whitney U tests with Holm correction. A check is counted as a nominally significant risk-layering check when the expected risk order holds, the Kruskal–Wallis p -value is below 0.05, and all three adjusted pairwise tests support bear > neutral, neutral > bull, and bear > bull. On the BTCUSDT boundary benchmark, we additionally report a block-bootstrap robustness variant in which the three directional pairwise separations are rechecked with one-sided block-bootstrap resampling, and a block-size sensitivity sweep over 12-hour, 24-hour, 48-hour, and 72-hour blocks. In threshold-sensitivity experiments, a check is excluded when a state is absent in the corresponding asset–period slice; this mainly affects the strict rule on SOLUSDT and XRPUSDT and is reported through check-coverage flags.

These protocol choices reflect a trade-off. The seven risk metrics are fixed and reused unchanged across assets, threshold variants, and model families. They cover dispersion, absolute movement, asymmetric downside exposure, broader pathwise variation, and threshold-based loss incidence, which together provide a compact but multi-faceted view of ex-post risk. The three test subperiods balance two competing needs: enough observations per slice for non-parametric testing, and enough temporal resolution to reveal whether semantic orderings survive regime changes inside the 16-month test span. Holm correction is used because each check involves three directional pairwise comparisons; it controls the family-wise error rate while remaining less conservative than Bonferroni in this repeated, relatively low-sample setting. Because the observations remain serially dependent within market periods, the Kruskal–Wallis and Mann–Whitney p -values should be read as nominal diagnostics under an independence approximation, not as exact finite-sample evidence. The 24-hour bootstrap block size is retained as the main reference because it matches a single diurnal trading cycle for the 1-hour series, while the later 12/24/48/72-hour sweep checks whether the semantic ranking is overly sensitive to that choice. The supplementary block-bootstrap variant does not eliminate all dependence concerns, but it does provide a stricter robustness screen for whether the pairwise risk separations survive coarse serial resampling.

Metric	Definition	Higher means
future_vol_24	Realized volatility over the next 24 hours.	More future dispersion
future_abs_return_24	Absolute close-to-close return over the next 24 hours.	Larger future move
future_path_loss_24	Largest drawdown reached within the next 24 hours.	Deeper downside stress
future_path_gain_24	Largest upward excursion reached within the next 24 hours.	Larger intrahorizon upside swing
future_range_24	High-minus-low range realized within the next 24 hours.	Broader path variation
loss_hit_2pct_24	Indicator or rate for whether a 2% downside threshold is hit within 24 hours.	More frequent downside threshold hits
loss_hit_5pct_24	Indicator or rate for whether a 5% downside threshold is hit within 24 hours.	More severe downside threshold hits

6. Results

6.1. BTCUSDT Benchmark and Baseline Boundary

Table 1 establishes the central empirical tension of the paper: the model that best matches the proxy labels is not necessarily the one that produces the most credible state path. These classification scores measure agreement with the heuristic labels rather than direct regime accuracy, so they need to be read alongside switching behavior and the later semantic checks. Raw HistGradientBoosting nearly saturates heuristic-label replication at macro-F1 0.9945, yet it still switches 0.9011 times per day; that profile looks more like near-rule imitation than regime extraction. Once simple temporal post-processing is added, HistGradientBoosting remains the strongest label-fitting learned baseline, with macro-F1 0.8731, a daily switching rate of 0.4084, and an average state duration of 58.47 hours. The post-processed linear baseline is weaker on label fit, but remains informative because it is cheaper and more transparent. The neural models sharpen the trade-off. Default GRU and vanilla LSTM are smoother than the raw static models, yet both stay below the post-processed static baselines on proxy-label classification. TCN-96x4 reaches macro-F1 0.6817 and moves ahead of the default GRU, but it does so with substantially more switching. The lightweight Transformer stress test does not reverse this pattern either: it reaches macro-F1 0.5980 with daily switching rate 0.5474 and particularly weak bear recall. Even before semantics enter the discussion, fit, smoothness, semantic robustness, and model complexity are already pulling in different directions.

The Gaussian HMM is most informative as a boundary case. Under the default training-split majority-voting map, all three BTCUSDT hidden states collapse to the neutral label on the test split, with 7006, 2333, and 2062 observations assigned to neutral from the three latent partitions. That is why the reported bull and bear F1 scores are both zero even though the HMM still internally produces three latent-state IDs. This collapse should not be over-read as an intrinsic

Table 1

Condensed BTCUSDT benchmark. Best learned value per column is shown in bold. Transformer-2L is included only as a BTCUSDT stress-test reference. The Gaussian HMM row reports the proxy-majority-mapped path; the later semantic-remapping diagnostic uses a different ex-post state ordering, for which switching rises to 2.68/day, average duration falls below 9 hours, and nominal significant-risk checks reach 12/21.

Method	Macro-F1	Bull F1	Bear F1	Switch/day	Avg. duration	Sig. risk checks
Logistic regression	0.7697	0.6713	0.7351	1.2400	19.32	15/21 (71.4%)
Logreg + post-proc.	0.7435	0.6356	0.7015	0.5474	43.68	14/21 (66.7%)
HistGB	0.9945	0.9973	0.9879	0.9011	26.58	—
HistGB + post-proc.	0.8731	0.8186	0.8395	0.4084	58.47	10/21 (47.6%)
Gaussian HMM	0.2990	0.0000	0.0000	0.0000	11401.00	—
Main GRU	0.6651	0.6251	0.4598	0.5221	45.79	16/21 (76.2%)
Vanilla LSTM	0.6314	0.5338	0.4602	0.5179	46.16	12/21 (57.1%)
TCN-96x4	0.6817	0.6327	0.4975	0.6779	35.30	10/21 (47.6%)
Transformer-2L	0.5980	0.6338	0.2536	0.5474	43.68	12/21 (57.1%)

failure of latent-state modelling: under highly imbalanced weak labels, a training-period majority map is itself fragile under concept drift. A fairer ex-post semantic remapping therefore changes the picture, but not the practical conclusion. When the hidden states are reordered by their realized test-set risk profile, the HMM recovers 14/21 raw risk-order checks and 12/21 nominal significant-risk checks. However, that semantic rescue comes with a daily switching rate of 2.68 and average state duration below 9 hours, far worse than the practical learned baselines. The issue is therefore not simply that HMMs are unusable, but that in the present weak-label setting their label-fit mapping is unstable under concept drift, while the semantically reordered path is too switch-heavy to serve as a credible deployment reference.

The five-seed checks sharpen rather than overturn this ordering. Across five seeds, GRU reaches macro-F1 0.6637 ± 0.0120 , daily switching rate 0.5272 ± 0.0081 , risk-order rate 0.8857 ± 0.0261 , and nominal significant-risk rate 0.6381 ± 0.1287 . TCN-96x4 raises macro-F1 to 0.6819 ± 0.0126 , but does so with more switching at 0.7309 ± 0.0527 and a lower nominal significant-risk rate of 0.5714 ± 0.0583 . Transformer-2L is weaker on both classification and semantics, with macro-F1 0.5668 ± 0.0425 and significant-risk rate 0.4524 ± 0.0991 . The important point is therefore not that one temporal backbone dominates unconditionally, but that the fit-first ranking remains different from the semantic ranking even after repeated retraining. The ordering is robust enough to sustain the fit-versus-semantics tension, but not strong enough to justify a universally dominant temporal backbone claim. A similar signal appears on the static side in ETHUSDT: raw HistGradientBoosting reaches macro-F1 0.9942, its post-processed version reaches 0.8648, and post-processed logistic regression reaches 0.7391. The stronger non-linear baseline is therefore not merely a BTC-specific curiosity.

The temporal-backbone comparison also shows that the TCN story is not monotonic. The first run, TCN-64x3, improves over LSTM but not over GRU; the wider 128x3 variant then underperforms both; only the deeper 96x4 variant becomes the strongest neural classifier. The lightweight Transformer stress test lands closer to the recurrent models in switching behavior, but not in classification quality. That

matters. It suggests that simply changing backbone family is not automatically beneficial: a deeper TCN improves proxy-label fit, whereas the lightweight Transformer does not; neither change removes the need for semantic auditing. One plausible reason is architectural rather than accidental. On low-signal, non-stationary 1-hour crypto series, a small vanilla attention model without patching or channel-aware structure can overreact to local pseudo-correlations while failing to stabilize sparse bear/bull states. This should therefore be read as a limitation of the simple attention reference used here, not as a blanket claim against modern patch-based or channel-aware time-series Transformers. The trade-off persists, however: all tested non-GRU temporal alternatives give up at least one of classification quality, path smoothness, or significant risk layering.

6.2. Extended-Pool Tradeoff: Fit, Smoothness, and Semantics Disagree

Among the tested TCN variants, TCN-96x4 gives the highest macro-F1, so it remains the strongest temporal challenger to the default GRU. The broader point, however, is no longer just a GRU-versus-TCN comparison. Once the BTCUSDT candidate pool is expanded to include the post-processed logistic baseline, the post-processed HistGradientBoosting baseline, and the lightweight Transformer stress-test reference, no single model remains best on all deployment-relevant axes. HistGB with post-processing dominates proxy-label fit and also gives the smoothest learned state path among the practical learned baselines. The post-processed logistic model gives the highest raw risk-order consistency. The default GRU gives the highest nominal significant-risk-layering rate. TCN-96x4 remains the strongest temporal classifier on macro-F1 but gives up both smoothness and semantic layering, while the Transformer stress test stays in the interior of the trade-off and does not overturn the broader ranking-reversal story. Under the present protocol, the key result is therefore not a universal winner but a visibly split frontier.

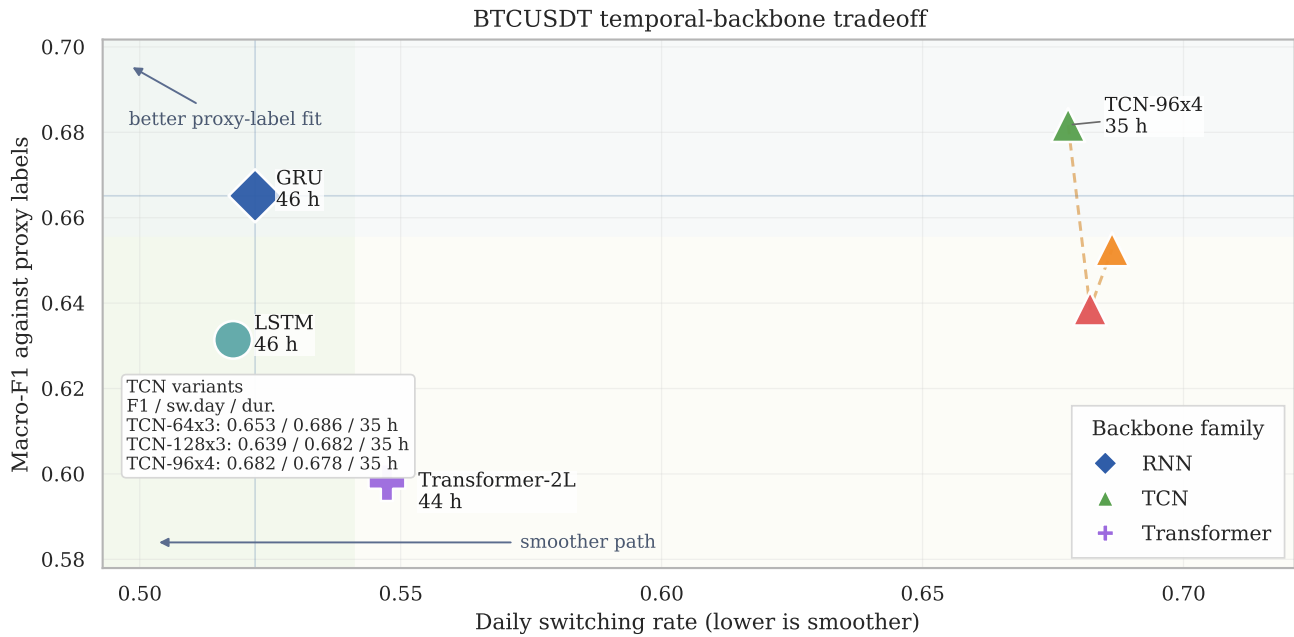


Figure 2: BTCUSDT temporal-backbone stress test. TCN-96x4 improves proxy-label macro-F1 over the default GRU, but all tested TCN variants have higher daily switching rates. Point labels also report average state duration over hours, so path persistence can be read directly rather than only through bubble area. The lightweight Transformer stress test stays closer to the recurrent models in switching behavior, yet still falls below GRU on both proxy-label fit and nominal significant risk layering. This result motivates evaluating state paths beyond proxy-label fit alone.

6.3. Risk-State Semantics Are Stronger Than Return Semantics

If the analysis stopped with the BTCUSDT benchmark, the practical verdict would look simple: HistGradientBoosting with post-processing would be the strongest static learner, the post-processed logistic model the simplest transparent comparator, and TCN-96x4 the strongest neural classifier. The three-dimensional protocol makes that summary incomplete. Once ex-post semantics are added, the question is no longer just which method best reproduces heuristic labels, but which one produces state paths whose downstream meaning survives changes in period, asset, and significance criterion.

Overall BTCUSDT state semantics show that the three-state taxonomy organizes future risk more consistently than future direction. Across the proxy labels, the most interpretable low-cost baseline (logistic regression with temporal post-processing), and the main GRU, the cleanest separation appears on future volatility: the historical bear state is highest risk, bull is lowest risk, and neutral lies between them. The return side is much less stable. In the proxy labels, and at times in the learned models as well, the historical bear state can still be followed by positive average returns.

Figure 4 makes the asymmetry visually explicit. Once the BTCUSDT test period is split into 2025-H1, 2025-H2, and 2026-YTD, the ordering of bear, neutral, and bull on future returns changes across subperiods, whereas the ordering on future volatility remains much more stable.

The holdout period mixes a rebound phase with two later drawdown phases, so the historical bear label often behaves more like a post-selloff high-risk state than like a persistent negative-return forecast. That is why every BTC learned method listed in Table 2, including the stronger HistGradientBoosting baseline and the lightweight Transformer stress test, ends up at 0/6 overall return-order consistency checks even while several methods retain strong raw or significant risk ordering.

The five-asset comparison in Table 2 makes the same point in a broader setting. The first semantic column reports raw risk-order consistency, the second reports return-order consistency, and the third reports the fraction of 21 risk checks that remain nominally significant under the nonparametric protocol. Notably, risk semantics are much stronger than return semantics across most asset–method pairs. The rankings, however, refuse to settle into a single order: on BTCUSDT, the post-processed logistic baseline has the best raw risk-order consistency, HistGradientBoosting with post-processing dominates proxy-label fit, TCN-96x4 has the best neural proxy-label classification score, the lightweight Transformer stress test sits between the weaker temporal baselines and the stronger static baselines, and the GRU records more nominally significant risk-layering checks than the other tested temporal backbones. The expanded assets offer both support and a warning. BNBUSDT and SOLUSDT preserve strong risk layering for the lightweight learned baselines, whereas XRPUSDT is materially weaker, especially for direct BTC-trained GRU transfer. The XRP

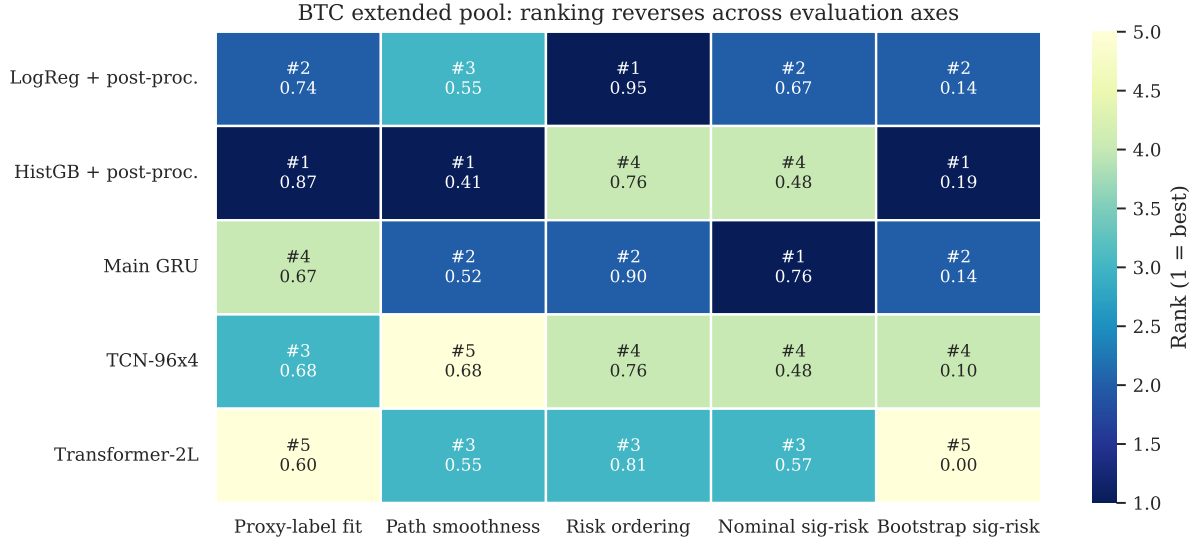


Figure 3: BTCUSDT extended-pool ranking reversal. Each column reports the within-pool rank on one evaluation axis, with rank 1 indicating the strongest performer and darker shading indicating a better rank. The winner changes across proxy-label fit, path smoothness, raw risk ordering, nominal significant risk layering, and bootstrap-robust significant risk layering. The bootstrap column reports the 24-hour block-bootstrap significant-risk-layering rate. HistGB + post-processing leads on fit and smoothness, LogReg + post-processing leads on raw risk ordering, and the main GRU leads on nominal significance. The figure makes explicit that no single model dominates once semantic diagnostics are added.

weakness is not only a transfer problem: even the XRP proxy labels themselves reach just 57.1% raw risk-order consistency and 5/21 nominal significant checks, suggesting that the underlying heuristic taxonomy is already much noisier on that asset. The bootstrap uncertainty intervals reported in the supplementary experiment tables should therefore be read cautiously; they are better viewed as robustness intervals than as high-precision estimates because each rate is based on a small number of period–metric checks.

The block-bootstrap check gives a more conservative result, and the contraction is not limited to one arbitrary block choice. In the current BTCUSDT sensitivity sweep, the main GRU falls from 16/21 nominal significant checks to 6/21 under 12-hour blocks and to 3/21 under 24-hour, 48-hour, and 72-hour blocks. LogReg + post-processing falls from 14/21 to 7/21, then 3/21, 2/21, and 1/21 as the blocks lengthen. Proxy labels, HistGB + post-processing, and TCN-96x4 show the same basic pattern: once serial dependence is respected more aggressively, the nominal significance gaps compress and the stronger BTC candidates become harder to separate cleanly. The practical implication is not that one model retains an unquestionable dependence-aware semantic lead, but that naive independence-style significance screens can overstate between-model separation under heavily overlapping windows. These results indicate that the nominal counts are useful for ranking models within the protocol, although they should not be treated as high-confidence finite-sample evidence. The bootstrap result is therefore best read as a dependence-aware stress test rather than a search for one uniquely correct block size.

6.4. Increment Over Conventional Multi-Criteria Evaluation

The preceding results motivate a more explicit comparison with existing multi-criteria evaluation practice. Once different methods lead on different axes, a conventional MCDM pipeline will usually compress that disagreement into a single score and may thereby hide the semantic loss induced by the chosen weighting rule. To quantify that risk, we compare the proposed semantic protocol against four selection rules. One uses classification-only selection by proxy-label macro-F1. Another uses a conventional weighted multi-criteria score over classification and stability, with normalized macro-F1, balanced accuracy, and inverse daily switching rate weighted 0.4, 0.3, and 0.3. A third applies TOPSIS to the same classification-plus-stability criterion set. The last is an oracle-style weighted score that also includes nominal significant-risk layering; it is included to test whether merely adding a semantic scalar is enough, as opposed to treating semantics as a separate diagnostic axis. For this comparison only, the protocol is operationalized as a semantic selector that first maximizes nominal significant-risk layering, then uses lower switching and higher macro-F1 as tie-breakers. This selector is not meant to replace the paper’s multi-objective reporting; it is a device for measuring selection disagreement and semantic regret, defined here as the selected method’s lost nominal significant-risk-layering rate relative to the protocol selector. The point is therefore not that TOPSIS or weighted scoring are wrong in general, but that they answer a different question: they optimize over chosen observables, whereas the

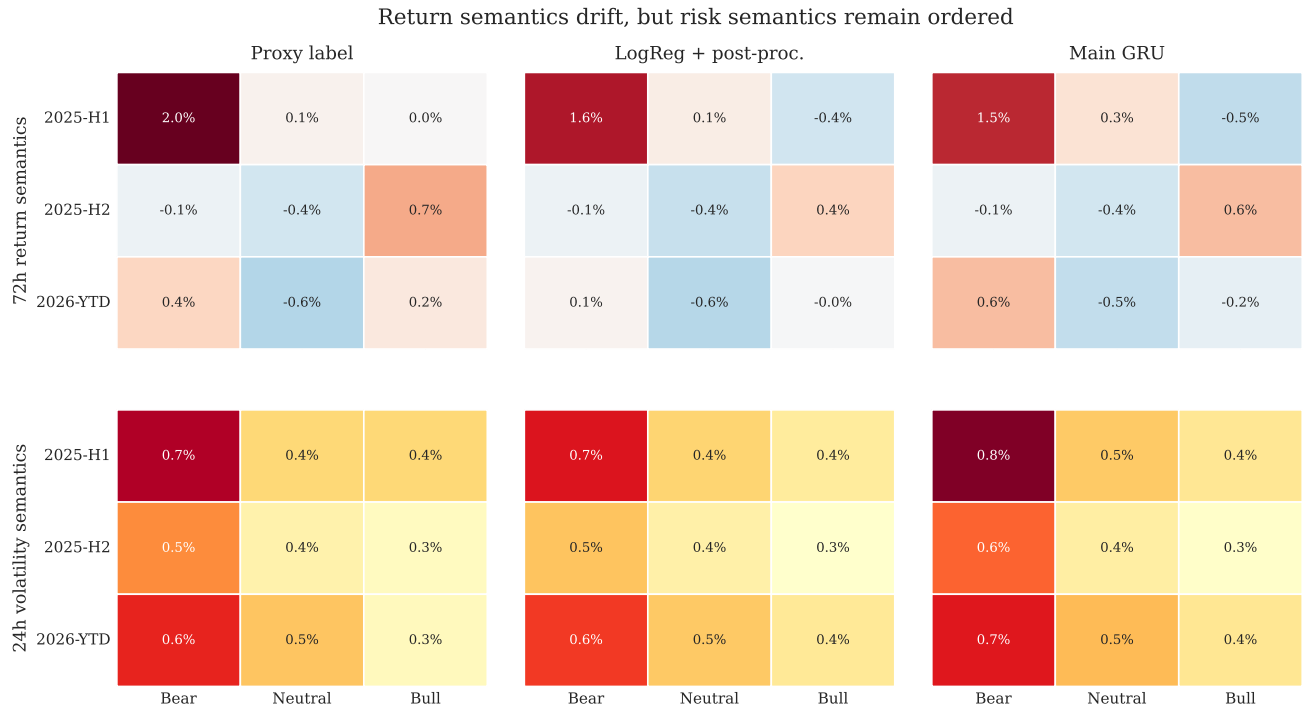


Figure 4: BTCUSDT return semantics are unstable, whereas risk semantics are comparatively stable. The top row shows 72-hour directional-return semantics for the proxy labels, LogReg + post-processing, and the main GRU; the ordering of bear, neutral, and bull shifts across 2025-H1, 2025-H2, and 2026-YTD. The bottom row shows 24-hour volatility semantics for the same three state paths; here the bear/high-risk state remains highest and the bull/low-risk state remains lowest much more consistently. The return and volatility rows use independent color scales; warmer colors indicate higher values within each row, and cross-row color comparisons are not meaningful.

Table 2

Three-dimensional semantic evaluation. Risk-order consistency counts how often the expected order bear > neutral > bull appears across 21 risk checks; return-order consistency counts how often bull > neutral > bear appears across six return checks; significant risk checks count how often the expected risk order is also supported by Kruskal–Wallis and Holm-corrected one-sided Mann–Whitney tests under the nominal independence approximation. Within each asset block, the strongest value on each semantic axis is shown in bold. BTCUSDT block-bootstrap robustness and block-size sensitivity are reported separately in Figure 5.

Asset	Method	Risk-order consistency	Return-order consistency	Sig. risk checks
BTC	Proxy label	81.0%	0.0%	11/21 (52.4%)
BTC	Logistic regression	90.5%	0.0%	15/21 (71.4%)
BTC	Logreg + post-proc.	95.2%	0.0%	14/21 (66.7%)
BTC	HistGB + post-proc.	76.2%	0.0%	10/21 (47.6%)
BTC	Main GRU	90.5%	0.0%	16/21 (76.2%)
BTC	Vanilla LSTM	71.4%	0.0%	12/21 (57.1%)
BTC	TCN-96x4	76.2%	0.0%	10/21 (47.6%)
BTC	Transformer-2L	81.0%	0.0%	12/21 (57.1%)
ETH	Proxy label	66.7%	33.3%	9/21 (42.9%)
ETH	Logistic regression	81.0%	33.3%	12/21 (57.1%)
ETH	Logreg + post-proc.	76.2%	50.0%	13/21 (61.9%)
ETH	Main GRU (ETH-trained)	71.4%	16.7%	10/21 (47.6%)
ETH	BTC-trained GRU	76.2%	0.0%	9/21 (42.9%)
BNB	Proxy label	66.7%	66.7%	12/21 (57.1%)
BNB	Logistic regression	95.2%	33.3%	15/21 (71.4%)
BNB	Logreg + post-proc.	95.2%	33.3%	17/21 (81.0%)
BNB	BTC-trained GRU	57.1%	0.0%	7/21 (33.3%)
SOL	Proxy label	66.7%	0.0%	9/21 (42.9%)
SOL	Logistic regression	90.5%	0.0%	14/21 (66.7%)
SOL	Logreg + post-proc.	90.5%	0.0%	14/21 (66.7%)
SOL	BTC-trained GRU	76.2%	16.7%	13/21 (61.9%)
XRP	Proxy label	57.1%	0.0%	5/21 (23.8%)
XRP	Logistic regression	61.9%	0.0%	6/21 (28.6%)
XRP	Logreg + post-proc.	71.4%	0.0%	6/21 (28.6%)
XRP	BTC-trained GRU	23.8%	0.0%	0/21 (0.0%)

BTCUSDT: significance shrinks under longer bootstrap blocks

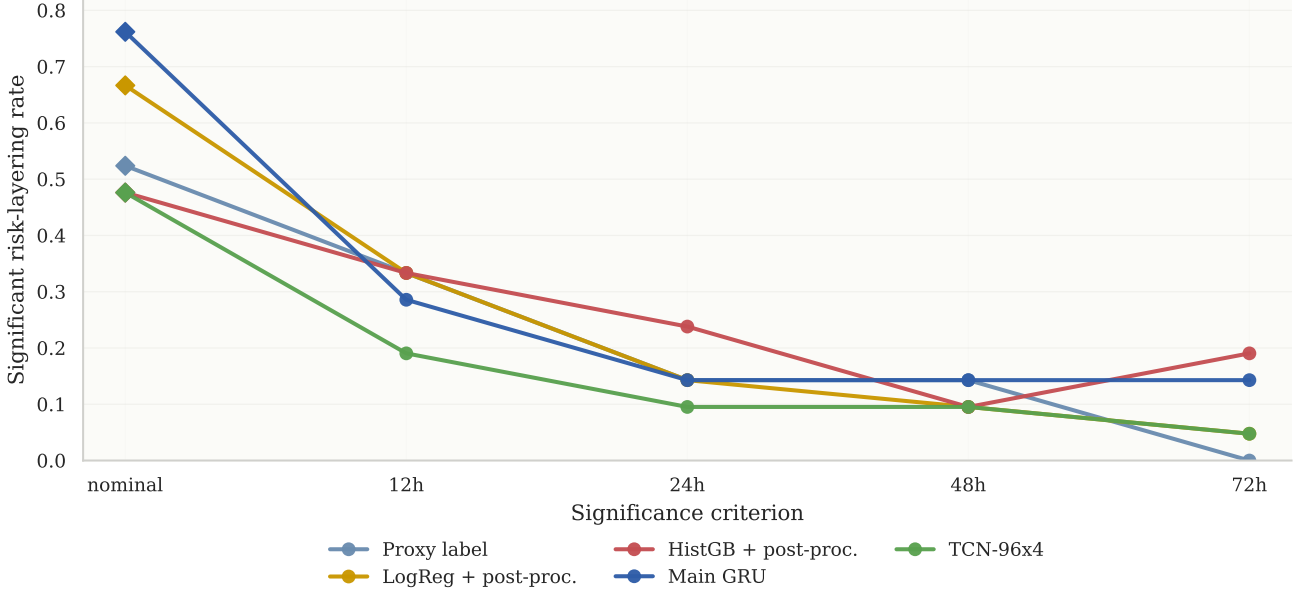


Figure 5: BTCUSDT block-bootstrap sensitivity by block size. Each slope traces the significant-risk-layering rate from the nominal screen to 12-hour, 24-hour, 48-hour, and 72-hour bootstrap blocks. All methods contract materially, and the between-method gaps compress as the blocks lengthen, which shows that independence-style significance screens can make the semantic ranking look more decisive than it remains under dependence-aware resampling.

audit protocol asks whether those observables are sufficient once semantic validity is inspected directly.

Definition: Semantic regret. For a given model-selection rule and candidate pool, semantic regret is the loss in nominal significant-risk-layering rate incurred by the selected model relative to the best-performing model on that semantic axis. It quantifies the hidden deployment cost of favoring proxy-label fit or stability when those criteria point to a model with weaker downstream risk semantics. In practical terms, it measures how much state meaning is lost when model choice is compressed too early into a single score.

Two candidate pools help clarify where the protocol actually adds something. The five-asset common pool contains logistic regression, logistic regression with post-processing, and the relevant GRU line on each asset. In that pool, classification-only selection disagrees with the semantic protocol on every asset because it always favors the raw logistic model, despite its much higher switching. Once stability is added, most of that discrepancy disappears: weighted scoring and TOPSIS disagree with the protocol on only 20% of assets, with a mean semantic regret of 0.019 significant-risk rate, or about 0.4 of 21 checks. When the candidate set is simple, the increment over standard classification-plus-stability MCDM is fairly modest.

The BTCUSDT extended-pool comparison summarized in Table 3 gives a different result. Once the stronger Hist-GradientBoosting and TCN candidates enter, classification-only, weighted classification-plus-stability, and TOPSIS all select HistGB with post-processing because it combines

the highest macro-F1 (0.8731) with the lowest switching rate (0.4084). Even the oracle-style scalar that includes a semantic term does not recover the GRU; it selects the post-processed logistic baseline and still loses 2 of 21 significant-risk checks. In this setting, a method can look reasonable under conventional MCDM and still miss the stronger ex-post risk semantics on the nominal screen. This is where the protocol becomes most consequential. The later bootstrap contraction is important here: it does not erase the selection disagreement, but it does show that the protocol's added value lies more in exposing ranking fragility than in certifying a single dependence-robust winner.

6.5. Checkpoint Selection as an Audit Problem

The same tension appears one step earlier, at checkpoint selection time. Up to this point, all temporal models have still used the best validation macro-F1 checkpoint. That is a sensible default for supervised sequence classification, but it is not obviously aligned with the paper's main claim that proxy-label fit is an incomplete deployment criterion. We therefore ran a focused diagnostic on the BTCUSDT GRU. The first selector retained the checkpoint with the highest validation macro-F1. The second used a semantic-aware lexicographic rule on the validation split: it prioritized nominal significant-risk-layering rate, then risk-order rate, then validation macro-F1, and only then lower daily switching rate. A stricter significance-first rule was also tried and immediately selected an immature epoch with much worse test classification, so the reported semantic selector

Table 3

Increment over conventional multi-criteria evaluation. Disagreement is the fraction of assets in the pool where the framework selects a different method from the semantic protocol selector. Semantic regret is the selected method's lost nominal significant-risk-layering rate relative to the protocol selector; 0.2857 corresponds to 6/21 checks. Positive macro-F1 cost means the conventional framework selected a model with higher proxy-label macro-F1 than the protocol selector. Switch cost is the selected method's daily switching rate minus the protocol selector's rate.

Candidate pool	Framework	Disagreement	Semantic regret	Macro-F1 cost	Switch cost
Five-asset common	C only	100%	0.038	0.044	0.898
Five-asset common	C+T weighted	20%	0.019	0.016	0.005
Five-asset common	C+T TOPSIS	20%	0.019	0.016	0.005
Five-asset common	C+T+S weighted	20%	0.010	0.021	0.144
BTC extended	C only	100%	0.286	0.208	-0.114
BTC extended	C+T weighted	100%	0.286	0.208	-0.114
BTC extended	C+T TOPSIS	100%	0.286	0.208	-0.114
BTC extended	C+T+S weighted	100%	0.095	0.078	0.025

Table 4

Checkpoint-selection diagnostic for the BTCUSDT GRU. Main window uses the paper's default split (train 2021–2023, validation 2024, test 2025-01 to 2026-04). Rolling window uses train 2021–2022, validation 2023, and test 2024. Significant risk checks report nominal significant-risk layering counts under the same protocol used elsewhere in the paper. Check coverage can differ between selectors when state sparsity makes some rolling-window checks unevaluable, so percentages should be interpreted together with their denominators.

Window	Selector	Best epoch	Test macro-F1	Test bal. acc.	Sig. risk checks	Switch/day
Main	Validation macro-F1	12	0.6635	0.6533	14/21 (66.7%)	0.5053
Main	Semantic-aware	9	0.6304	0.6245	16/21 (76.2%)	0.5053
Rolling 2024	Validation macro-F1	12	0.5720	0.5340	1/14 (7.1%)	0.5247
Rolling 2024	Semantic-aware	8	0.5028	0.4983	1/7 (14.3%)	0.4427

keeps semantic criteria first while still requiring a reasonably mature classifier.

Table 4 shows that semantic-aware checkpointing is neither vacuous nor universally dominant. On the main BTCUSDT window, it changes the selected checkpoint from epoch 12 to epoch 9. That lowers test macro-F1 from 0.6635 to 0.6304, but it raises nominal significant risk layering from 14/21 to 16/21 without increasing daily switching. In other words, the validation split contains enough semantic signal for checkpoint selection to shift the deployment trade-off in the same direction as the paper's broader audit argument.

The rolling-origin window gives a more cautious result. When 2023 is used as validation and 2024 as test, semantic evidence on the validation split is much weaker: the semantic selector never observes a positive nominal significant-risk-layering rate and is forced to rely on weaker ordering signals. In that setting, semantic-aware checkpointing still produces a smoother test path and a higher significant-risk-layering rate under reduced check coverage, but the classification cost is larger and the gain in ex-post semantics remains modest. The practical implication is that semantic-aware checkpointing should itself be audited. It is most useful when the validation window already contains a detectable risk-state structure; when that structure is weak, semantic checkpointing becomes a conservative bias rather than a reliable improvement rule.

6.6. Threshold Sensitivity and Ablation Checks

The threshold sweep tests whether the main semantic pattern is merely an artifact of one hand-tuned rule. It

does not appear to be. Across all five assets, loosening or tightening the return, EMA-gap, and drawdown thresholds leaves cross-asset mean risk-order rates substantially higher than mean return-order rates. The strict rule also exposes a practical limit of hand-built taxonomies. On SOLUSDT and XRPUSDT, it leaves only 17 and 14 test bull bars, respectively, which reduces some risk checks from 21 to 7 or 14. This result also indicates that pushing the heuristic labels toward apparent precision can easily destroy state coverage.

Changing the sampling frequency gives a related robustness signal. On BTCUSDT at 4-hour bars, all four lightweight baselines still produce 0/6 return-order matches, while the corresponding risk-order rates remain between 9/14 and 10/14 and significant-risk counts remain between 4/14 and 6/14. The pattern therefore weakens but does not reverse when the bar frequency changes. ETHUSDT is more diagnostic than confirmatory in this test, because the 4-hour proxy labels become too sparse on the bull side and the test split effectively loses that state.

The drawdown deconfounding check is especially revealing because `max_drawdown_72` sits in both the feature set and the weak-label rule. That creates a legitimate concern: perhaps the recovered risk semantics are partly inherited from a hand-coded downside-stress anchor. Removing the feature weakens both shallow and deep models, which indicates that it carries real signal. But the magnitude of the weakening is uneven. Logistic regression drops from macro-F1 0.7697 to 0.7051 and from 71.4% to 61.9% in significant risk layering; the GRU drops from macro-F1 0.6651 to 0.6223 and from 76.2% to 38.1%. This pattern suggests that

Table 5

Threshold-sensitivity summary across five assets. Values are cross-asset means for each variant–method pair. The strict variant has incomplete check coverage for some SOLUSDT/XRPUSDT rows because the bull state becomes nearly absent.

Variant	Method	Mean risk-order rate	Mean sig. risk rate	Mean return-order rate	Min risk checks
Default	Proxy label	67.6%	43.8%	20.0%	21
Default	Logistic regression	83.8%	59.0%	13.3%	21
Default	Logreg + post-proc.	85.7%	61.0%	16.7%	21
Loose	Proxy label	69.5%	41.9%	23.3%	21
Loose	Logistic regression	75.2%	49.5%	10.0%	21
Loose	Logreg + post-proc.	75.2%	57.1%	16.7%	21
Strict	Proxy label	68.6%	46.7%	13.3%	7
Strict	Logistic regression	69.0%	54.8%	13.3%	14
Strict	Logreg + post-proc.	68.1%	50.0%	16.7%	14

Drawdown deconfounding: label fit can rise while semantic validity collapses

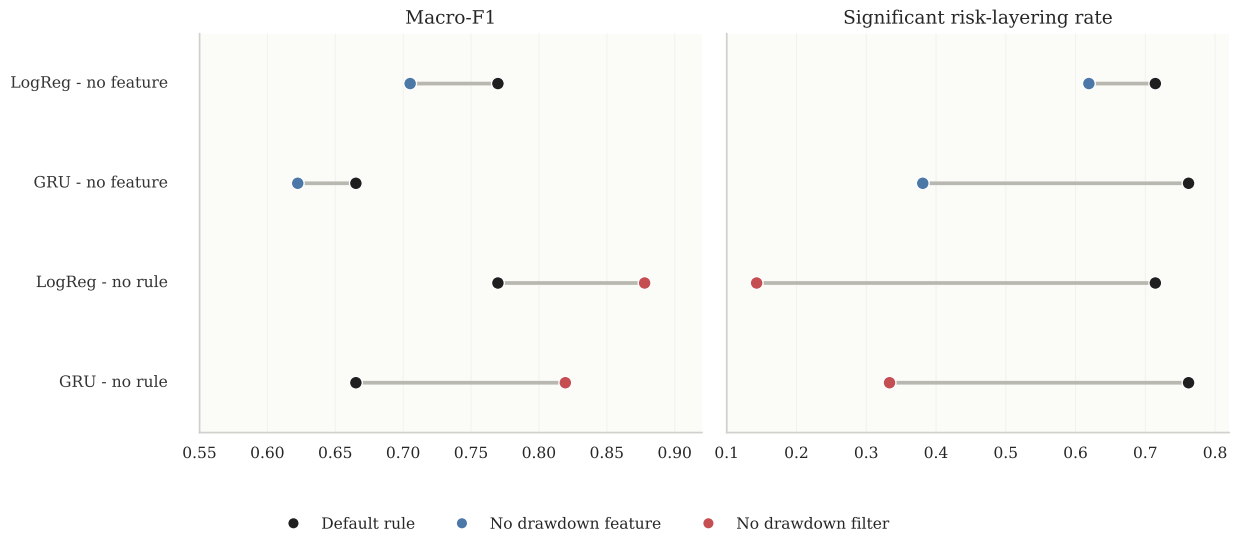


Figure 6: BTCUSDT drawdown deconfounding. Each dumbbell compares the default setup against a drawdown-removed variant for the same model. Removing the feature lowers both macro-F1 and significant risk layering, whereas removing the drawdown filter from the weak-label rule increases macro-F1 while sharply reducing significant risk layering. The figure makes explicit that drawdown is a transparent modelling anchor rather than a neutral ingredient.

the current GRU leans more heavily on the drawdown feature than the shallow baseline does. That should not be overstated, because the no-drawdown logistic model still retains substantial risk layering. Even so, the drawdown anchor is better treated as an important and transparent design choice than as an incidental feature.

The complementary rule-side deconfounding check shows the same pattern more clearly. Once the drawdown thresholds are removed from the weak-label rule, classification becomes easier while semantics deteriorate. Logistic regression improves from macro-F1 0.7697 to 0.8778, and the GRU improves from 0.6651 to 0.8194; at the same time, significant risk-layering counts fall from 15/21 to 3/21 for logistic regression and from 16/21 to 7/21 for the GRU. Even the drawdown-free proxy labels themselves retain only 3/21 significant risk-layering checks. This contrast suggests that removing the drawdown anchor does not destroy learnability, but it does weaken the cleanest ex-post risk separation.

The present state taxonomy is therefore partly anchored by drawdown-based weak supervision, although this again should be interpreted cautiously rather than as a claim that the entire result is an artifact.

The auxiliary-loss controls suggest a more modest interpretation. Removing $\mathcal{L}_{\text{bal}}$ changes the BTCUSDT GRU test macro-F1 from 0.6651 to 0.6628 and the daily switching rate from 0.5221 to 0.5053, while preserving the same significant risk-layering count as the main GRU (16/21). Removing the continuity term leaves macro-F1 almost unchanged at 0.6671 and again preserves 16/21 significant risk-layering checks. Removing the volatility gate changes little as well: macro-F1 0.6655 and 16/21 significant risk-layering checks. The stride-48 no-continuity control is much more damaging: once overlap support is sharply reduced, the BTCUSDT test set shrinks to 238 windows, macro-F1 falls to 0.5794, daily switching rate falls to 0.2996 simply because long overlapping runs are absent, and the significant risk-layering

Table 6

BTCUSDT label-side drawdown-rule deconfounding. Removing the drawdown thresholds from the weak-label rule improves proxy-label fit for both learned models, but it substantially weakens statistically significant risk layering.

Weak-label rule	Model	Macro-F1	Sig. risk checks
Default	Logistic regression	0.7697	15/21 (71.4%)
Default	Main GRU	0.6651	16/21 (76.2%)
No drawdown filter	Logistic regression	0.8778	3/21 (14.3%)
No drawdown filter	Main GRU	0.8194	7/21 (33.3%)

rate collapses to 0/21. This pattern suggests that, under the present setup, the continuity regularizer itself adds only a modest increment on top of highly overlapping windows, whereas overlap reduction substantially weakens the semantic structure recovered by the model. The results therefore do not point to any single auxiliary term as decisive. Instead, they place the temporal objective as one configuration inside a broader protocol whose value is better judged by stability and semantic diagnostics than by a large isolated component effect. This raises a sharper question: if no single auxiliary term is decisive, can semantic validity be improved by directly targeting it as a training objective?

6.7. Information Boundaries: What Weak Supervision Can and Cannot Learn

The SAWS experiments provide a negative but informative result. Across all tested semantic-loss weights from 0.05 to 2.00, and under both the default high-overlap setting and the stride-48 low-overlap diagnostic, the tested semantic-regularization variants produce no measurable improvement in risk-layering checks. The training-time semantic gap remains near zero throughout, indicating that the model already satisfies approximate risk ordering without external guidance when that ordering is learnable at all. This suggests that semantic alignment, when it occurs, emerges primarily from the data structure presented by the proxy labels rather than from the added training objective itself.

To characterize that data structure, we compute a Semantic Orientation Ratio (SOR) for each asset–variant combination. SOR measures the ratio of mutual information between proxy labels and future 24-hour volatility to mutual information between proxy labels and future 24-hour return. Across the fifteen asset–variant combinations formed by five assets and three threshold variants, SOR is consistently high for BTC, ETH, BNB, and SOL (range 4.1–6.3), but substantially lower for XRP (range 1.2–3.6). Figure 7 shows the resulting separation directly: the XRP points cluster in the low-SOR, low-semantic-quality corner, whereas the remaining assets occupy a higher-SOR region with materially stronger nominal risk layering. This separation corresponds directly to the semantic-quality results in Table 2: XRP is the only asset where none of the tested learned models achieves strong risk layering. In that sense, SOR functions as an anomaly diagnostic. At the label-audit stage, before any downstream learner is trained on the proxy labels, it identifies assets whose proxy-label information structure is already misaligned with the paper’s risk-stratification objective.

That anomaly is also economically interpretable. During the 2021–2026 sample, XRP was repeatedly repriced by litigation and regulatory-news shocks around the SEC case, including swings in court-ruling expectations and exchange relisting sentiment, making its short-horizon state changes far more event-driven than those of BTC or ETH. In such an environment, realized regime switches need not align with the smoother volatility–drawdown cycles that underpin the proxy-label rule used here. The low SOR of XRP is therefore not just an information-theoretic curiosity. It is consistent with a market whose salient transitions are externally triggered and only weakly captured by a proxy taxonomy designed around persistent technical-risk signals.

The predictive scope of SOR is, however, bounded. When XRP is excluded, the correlation between SOR and downstream significant-risk-layering rate reverses sign rather than merely weakening. SOR therefore does not function as a continuous predictor of semantic quality across the remaining assets. Its diagnostic value is specific to identifying structural misalignment at the extremes rather than ranking assets within the normal range. This limitation matters because it prevents SOR from being overstated as a universal scalar quality index. Within the range of proxy-label designs that already carry enough risk-oriented information, semantic quality is shaped by factors other than information ratio alone, including temporal overlap structure, model capacity, and the presence or absence of the drawdown anchor.

Together, the SAWS and SOR results characterize the information boundaries of weak supervision in this setting. Explicit downside anchors can materially strengthen the supervisory signal, whereas removing those anchors increases label fit while sharply weakening ex-post risk layering. The practical implication is that later model-side interventions operate within, rather than override, the semantic information already encoded by proxy-label design.

7. Discussion

The central question posed in this paper is what weak supervision can learn from heuristic proxy labels, and what determines the boundary of that learning. The experiments give a consistent answer. Weak supervision recovers risk stratification reliably but not return direction, and this asymmetry persists across assets, threshold variants, temporal backbones, and evaluation criteria. More broadly, the results point to proxy-label information structure as the binding constraint: when proxy labels carry sufficient information about persistent risk signals relative to directional noise,

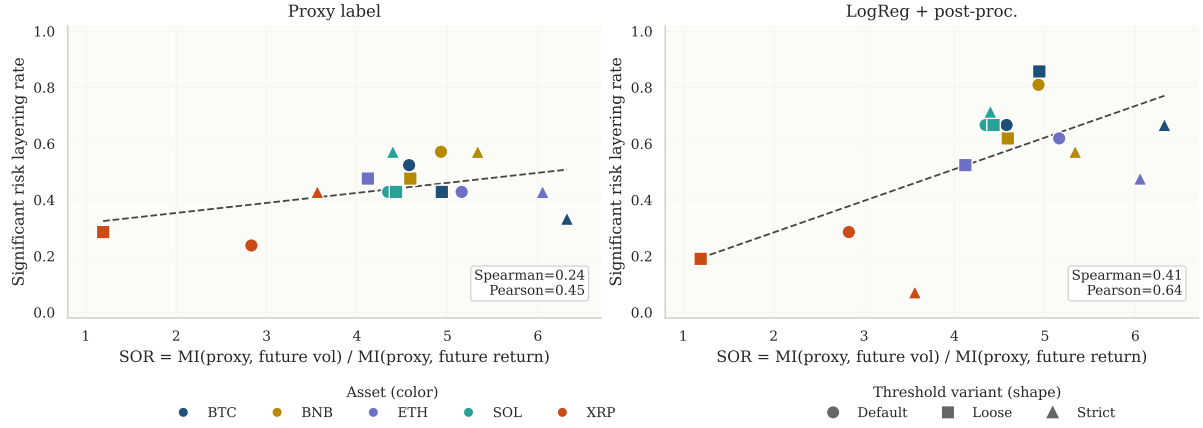


Figure 7: Semantic Orientation Ratio (SOR) as an anomaly diagnostic. Each point is one asset–threshold variant combination, with the same encoding in both panels: color denotes asset and marker shape denotes threshold variant. The left panel compares SOR against the proxy labels’ own significant-risk-layering rate; the right panel compares SOR against the downstream rate of LogReg + post-processing. XRP is the only asset that occupies the low-SOR, low-semantic-quality corner across all three threshold variants, while the remaining assets cluster in a higher-SOR region. The figure therefore supports SOR as a diagnostic for structural proxy-label misalignment, not as a continuous predictor within the normal asset range.

semantic alignment emerges naturally from training; when they do not, as in XRP, none of the tested downstream interventions recovers it clearly.

The ranking also depends sharply on the evaluation axis. Post-processed HistGradientBoosting is strongest on label fit, TCN-96x4 is the strongest tested neural classifier on that axis, and the default GRU looks more favorable mainly under the nominal significant-risk criterion on BTCUSDT. Even that advantage becomes less distinct once block-bootstrap checks are added. The main result is therefore not a universal winner but an empirical demonstration that weakly supervised state-learning methods can exchange places once the evaluation target shifts from label imitation to stability and semantics. The methodological contribution is to make that shift inspectable rather than implicit: the protocol exposes when a seemingly strong model owes its appeal mainly to heuristic-label fidelity while giving back too much in stability or semantic reliability. In that sense, the contribution is not another scalar selector layered on top of TOPSIS-style practice, but a transparent way to diagnose when scalarized selection starts to prefer semantically weaker state paths and when independence-style significance screens make that preference look more decisive than it is.

For knowledge-oriented financial monitoring systems, the weak labels in this study should not be read as the deployed knowledge itself, but as a domain-informed supervisory interface built from transparent market heuristics. The deployable object is the inferred state path and its downstream interpretation in risk monitoring, regime filtering, and decision support. Under that view, the protocol acts as an audit layer between model development and deployment: it helps reject candidates that score well against proxy labels yet encode fragile, unstable, or semantically unreliable market-state representations. The same logic extends beyond cryptocurrency data. Any weakly supervised state-learning

system that derives labels from imperfect operational heuristics faces the risk that learning to fit those heuristics will out-run learning to produce a trustworthy state representation.

Within that interpretation, continuity regularization, the balance penalty, and the volatility gate look more like secondary inductive biases than primary drivers of the result. Their estimated effects are modest under the tested settings. More revealing are the comparisons with post-processed logistic regression, HistGradientBoosting, the tuned TCN, and the checkpoint-selection diagnostic, because those comparisons expose how quickly the ranking can reverse once semantics are brought back in. The overlap controls sharpen that point further: the stride-48 no-continuity run suggests that heavily overlapping 48-hour windows already inject an implicit smoothness bias, so apparent path persistence should not automatically be credited to one auxiliary loss term. The classical HMM serves a different purpose. It mainly functions as a negative reference: under proxy-majority mapping its states collapse to neutral, whereas a semantic re-ordering can recover only partial risk ordering at the cost of an excessively switch-heavy path.

7.1. Design Implications

The results suggest five practical design implications for deployment-oriented financial monitoring systems.

- Proxy-label fidelity and downstream utility should be audited separately. Strong validation performance on heuristic labels is not a reliable proxy for whether the resulting state path will remain meaningful in use.
- Stronger temporal extractors are not automatically safer choices. In the present setup, the best TCN improves label fit while increasing semantic regret relative to the main GRU.

- Transparent modelling anchors matter. The drawdown checks show that explicit and interpretable risk anchors can materially shape whether the learned states remain semantically valid after training.
- Risk stratification is a more stable target than directional return forecasting. For the weak labels used here, the learned states carry more consistent information about future risk than about future direction, which should guide the design of downstream tasks.
- Proxy-label information structure is the binding constraint. In practice, semantic orientation should be checked before model development rather than relying on post-hoc auditing to rescue information-poor supervision.

7.2. Limitations

The present claims are bounded by several limitations. The labels are heuristic rather than ground truth, so proxy-label metrics are not direct measures of regime-identification accuracy. The cross-asset analysis covers only five liquid crypto spot pairs and does not extend to other asset classes or to richer inputs such as order-book, news, and sentiment data. The temporal baseline suite is also more extensive for BTCUSDT than for the added assets. In addition, the drawdown deconfounding checks show that one hand-crafted risk anchor materially affects both the labels and the learned models.

The significance analysis is also incomplete: the cross-asset table relies on nominal nonparametric diagnostics under an independence approximation, and the BTC block-bootstrap analysis shows that those counts can shrink sharply once serial dependence is partly respected. The threshold-sensitivity results likewise show that strict hand-crafted rules can make some states too sparse, especially the bull state for SOLUSDT and XRPUSDT, and the 4-hour ETH diagnostic becomes too sparse to support a balanced semantic test. The Transformer comparison is also deliberately limited: the paper includes only a lightweight vanilla attention baseline, not patch-based or channel-aware architectures such as stronger modern time-series Transformers. Checkpoint selection remains only partially resolved. The new diagnostic shows that semantic-aware checkpointing can improve test-time risk layering on the main BTCUSDT window, but it also shows that this benefit depends on the validation window itself containing a detectable semantic signal. The protocol should therefore be read as a compact benchmark and audit framework, not as a definitive market-state standard.

8. Conclusion

This paper characterizes the information boundaries of weak supervision in cryptocurrency market-state learning. The central finding is that heuristic proxy labels reliably transmit risk-stratification structure to learned representations, but not return-direction structure, and that this transmission is determined primarily by proxy-label information

content rather than by training objectives. The three-axis evaluation protocol proposed here makes that characterization operational: by separating proxy-label agreement, temporal stability, and ex-post semantics, it exposes the gap between what models learn from supervision and what they deliver in deployment. Just as importantly, the block-bootstrap stress test shows that nominal semantic gaps can contract materially under dependence-aware resampling, so the protocol's value is not to proclaim one definitive winner but to reveal where apparently strong conclusions are being propped up by fragile statistical assumptions.

The central design lesson is that fidelity to noisy labels and usefulness of the learned state path are not interchangeable objectives. Classification-plus-stability selection is often adequate in a simple candidate pool, but it can fail once stronger label-fitting models enter the comparison. In the extended BTCUSDT pool, it loses 6/21 nominal significant risk-layering checks relative to the semantic selector. The checkpoint-selection diagnostic shows that the same trade-off can appear even within one model family: on the main BTCUSDT window, a semantic-aware selector recovers 2 additional nominal significant risk checks without increasing switching, but on an earlier rolling-origin window the semantic signal is too weak for this rule to be reliably advantageous. For developers of crypto risk-monitoring, regime-filtering, and automated trading systems, the main implication is that proxy-label fit alone is an unsafe deployment criterion when the downstream state path must remain stable and interpretable, but semantic checkpointing should itself be conditioned on validation windows that actually exhibit stable risk-state structure. The broader lesson is that weakly supervised state-learning systems should be judged by how well they preserve stable and meaningful state representations, not only by how faithfully they reproduce the heuristic labels that trained them, and any such judgment should explicitly account for how serial dependence erodes overconfident semantic significance claims.

CRedit authorship contribution statement

Junwen Lou: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft. **Fukai Zhang:** Supervision, Methodology, Writing - review & editing. **Shuo Tian:** Supervision, Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The raw Binance spot candlestick data used in this study are publicly available from Binance. Processed datasets, code, experiment scripts, and generated outputs supporting this study are available upon reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used generative AI and AI-assisted technologies to improve language, readability, and manuscript organization. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] J. D. Hamilton, A new approach to the economic analysis of nonstationary time series and the business cycle, *Econometrica* 57 (2) (1989) 357–384. doi:10.2307/1912559.
- [2] L. R. Rabiner, A tutorial on hidden markov models and selected applications in speech recognition, *Proceedings of the IEEE* 77 (2) (1989) 257–286. doi:10.1109/5.18626.
- [3] A. Ang, A. Timmermann, Regime changes and financial markets, *Annual Review of Financial Economics* 4 (1) (2012) 313–337. doi:10.1146/annurev-financial-110311-101808.
- [4] R. S. Mamon, R. J. Elliott, *Hidden Markov Models in Finance*, International Series in Operations Research & Management Science, Springer, New York, NY, 2007. doi:10.1007/0-387-71163-5.
- [5] L. Oelschläger, T. Adam, Detecting bearish and bullish markets in financial time series using hierarchical hidden markov models, *Statistical Modelling* 23 (2) (2023) 107–126. doi:10.1177/1471082X211034048.
- [6] S. Wang, R. Gupta, Y.-J. Zhang, Bear, bull, sidewalk, and crash: The evolution of the US stock market using over a century of daily data, *Finance Research Letters* 43 (2021) 101998. doi:10.1016/j.fr.l.2021.101998.
- [7] H. Nyberg, Predicting bear and bull stock markets with dynamic binary time series models, *Journal of Banking & Finance* 37 (9) (2013) 3351–3363. doi:10.1016/j.jbankfin.2013.05.008.
- [8] A. Koukorinis, G. W. Peters, G. Germano, Generative-discriminative machine learning models for high-frequency financial regime classification, *Methodology and Computing in Applied Probability* 27 (2). doi:10.1007/s11009-025-10148-8.
- [9] G. Figà-Talamanca, S. Focardi, M. Patacca, Regime switches and commonalities of the cryptocurrencies asset class, *The North American Journal of Economics and Finance* 57 (2021) 101425. doi:10.1016/j.najef.2021.101425.
- [10] D. Ardia, K. Bluteau, M. Rüede, Regime changes in Bitcoin GARCH volatility dynamics, *Finance Research Letters* 29 (2019) 266–271. doi:10.1016/j.fr.l.2018.08.009.
- [11] C.-Y. Tan, Y.-B. Koh, K.-H. Ng, K.-H. Ng, Dynamic volatility modelling of Bitcoin using time-varying transition probability markov-switching GARCH model, *The North American Journal of Economics and Finance* 56 (2021) 101377. doi:10.1016/j.najef.2021.101377.
- [12] I. Agakishiev, W. K. Härdle, D. Becker, X. Zuo, Regime switching forecasting for cryptocurrencies, *Digital Finance* 7 (1) (2025) 107–131. doi:10.1007/s42521-024-00123-2.
- [13] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, C. Ré, Snorkel: rapid training data creation with weak supervision, *The VLDB Journal* 29 (2–3) (2020) 709–730. doi:10.1007/s00778-019-00552-1.
- [14] J. Zhang, C.-Y. Hsieh, Y. Yu, C. Zhang, A. Ratner, A survey on programmatic weak supervision, *arXiv preprint arXiv:2202.05433*. <https://arxiv.org/abs/2202.05433> (2022).
- [15] Z.-H. Zhou, A brief introduction to weakly supervised learning, *National Science Review* 5 (1) (2018) 44–53. doi:10.1093/nsr/nwx106.
- [16] N. Natarajan, I. S. Dhillon, P. K. Ravikumar, A. Tewari, Learning with noisy labels, in: *Advances in Neural Information Processing Systems* 26, 2013, pp. 1196–1204, https://proceedings.neurips.cc/paper_files/paper/2013/hash/3871bd64012152bf53df04b401193f-Abstract.html.
- [17] B. Frenay, M. Verleysen, Classification in the presence of label noise: A survey, *IEEE Transactions on Neural Networks and Learning Systems* 25 (5) (2014) 845–869. doi:10.1109/TNNLS.2013.2292894.
- [18] H. Song, M. Kim, D. Park, Y. Shin, J.-G. Lee, Learning from noisy labels with deep neural networks: A survey, *IEEE Transactions on Neural Networks and Learning Systems* 34 (11) (2023) 8135–8153. doi:10.1109/TNNLS.2022.3152527.
- [19] Y. El-Khatib, A. Hatemi-J, On a regime switching illiquid high volatile prediction model for cryptocurrencies, *Journal of Economic Studies* 51 (2) (2024) 485–498. doi:10.1108/JES-03-2023-0134.
- [20] P. Kayal, S. Dutta, Regime switching and causal network analysis of cryptocurrency volatility: evidence from pre-COVID and post-COVID analysis, *Digital Finance* 6 (2) (2024) 319–340. doi:10.1007/s42521-023-00104-x.
- [21] L. Gao, W. Ye, R. Guo, Jointly forecasting the value-at-risk and expected shortfall of Bitcoin with a regime-switching CAViaR model, *Finance Research Letters* 48 (2022) 102826. doi:10.1016/j.fr.l.2022.102826.
- [22] V. Troster, A. K. Tiwari, M. Shahbaz, D. N. Macedo, Bitcoin returns and risk: A general GARCH and GAS analysis, *Finance Research Letters* 30 (2019) 187–193. doi:10.1016/j.fr.l.2018.09.014.
- [23] N. Borri, Conditional tail-risk in cryptocurrency markets, *Journal of Empirical Finance* 50 (2019) 1–19. doi:10.1016/j.jempfin.2018.11.002.
- [24] D. Koutmos, Liquidity uncertainty and Bitcoin’s market microstructure, *Economics Letters* 172 (2018) 97–101. doi:10.1016/j.econlet.2018.08.041.
- [25] N. James, Evolutionary correlation, regime switching, spectral dynamics and optimal trading strategies for cryptocurrencies and equities, *Physica D: Nonlinear Phenomena* 434 (2022) 133262. doi:10.1016/j.physd.2022.133262.
- [26] A. Shakourloo, A. Azimli, Regime-switching in Bitcoin volatility under global uncertainty: Markov-switching GARCH and hidden Markov Copula approaches, *Research in International Business and Finance* 83 (2026) 103295. doi:10.1016/j.ribaf.2026.103295.
- [27] Y. Wang, J. Andreeva, G. M. Martin-Barragan, Forecasting cryptocurrencies volatility using statistical and machine learning methods: A comparative study, *International Review of Financial Analysis* 87 (2023) 102914. doi:10.1016/j.irfa.2023.102914.
- [28] Q. Huang, I. Sangiorgi, A. Urquhart, Identification of bitcoin volatility drivers using statistical and machine learning methods, *Journal of International Financial Markets, Institutions and Money* 92 (2024) 102064. doi:10.1016/j.intfin.2024.102064.
- [29] M. S. Ahmed, A. A. El-Masry, A. I. Al-Maghyreh, S. Kumar, Cryptocurrency volatility: A review, synthesis, and research agenda, *Research in International Business and Finance* 71 (2024) 102472. doi:10.1016/j.ribaf.2024.102472.
- [30] J. Hernández-González, I. Inza, J. A. Lozano, Weak supervision and other non-standard classification problems: A taxonomy, *Pattern Recognition Letters* 69 (2016) 49–55. doi:10.1016/j.patrec.2015.10.008.
- [31] F. Schwenker, E. Trentin, Pattern classification and clustering: A review of partially supervised learning approaches, *Pattern Recognition Letters* 37 (2014) 4–14. doi:10.1016/j.patrec.2013.10.017.
- [32] F. Tonolini, N. Aletras, Y. Jiao, G. Kazai, Robust weak supervision with variational auto-encoders, in: *Proceedings of the 40th International Conference on Machine Learning*, Vol. 202 of *Proceedings of Machine Learning Research*, PMLR, Honolulu, Hawaii, USA, 2023, pp. 34394–34408, <https://proceedings.mlr.press/v202/tonolini23a.html>.
- [33] L. Yang, T. Gao, W. Wei, M. Dai, C. Fang, J. Duan, Multi-task meta label correction for time series prediction, *Pattern Recognition* 150 (2024) 110319. doi:10.1016/j.patcog.2024.110319.
- [34] P. Nystrup, E. Lindström, H. Madsen, Learning hidden markov models with persistent states by penalizing jumps, *Expert Systems with Applications* 150 (2020) 113307. doi:10.1016/j.eswa.2020.113307.
- [35] A. Bemporad, V. Breschi, D. Piga, S. P. Boyd, Fitting jump models, *Automatica* 96 (2018) 11–21. doi:10.1016/j.automatica.2018.06.022.
- [36] I. E. Livieris, S. Stavroyiannis, L. Iliadis, P. Pintelas, Smoothing and stationarity enforcement framework for deep learning time-series forecasting, *Neural Computing and Applications* 33 (20) (2021) 14021–14035. doi:10.1007/s00521-021-06043-1.
- [37] O. Blümke, A structural hidden markov model for forecasting scenario probabilities for portfolio loan loss provisions, *Knowledge-Based Systems* 249 (2022) 108934. doi:10.1016/j.knosys.2022.108934.
- [38] C.-L. Hwang, K. Yoon, *Multiple Attribute Decision Making: Methods and Applications*, Springer, Berlin, Heidelberg, 1981. doi:10.1007/

978-3-642-48318-9.

- [39] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Computation* 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.
- [40] K. Cho, B. van Merriënboer, Ç. Gülçehre, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using RNN encoder–decoder for statistical machine translation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734. doi:10.3115/v1/D14-1179.
- [41] S. Bai, J. Z. Kolter, V. Koltun, An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, *arXiv preprint arXiv:1803.01271*. <https://arxiv.org/abs/1803.01271> (2018).
- [42] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, *International Conference on Learning Representations*, <https://openreview.net/forum?id=Jbdc0vT0co1> (2023).
- [43] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, iTransformer: Inverted transformers are effective for time series forecasting, *International Conference on Learning Representations*, <https://openreview.net/forum?id=JePfAI8fah> (2024).